

UNIVERSITY OF TECHNOLOGY - VNUHCM	HOMEWORK 1	Semester/Academic year	1 / 2026-2027
FACULTY OF COMPUTER SCIENCE AND ENGINEERING	Foundations, Evaluation, and Generalisation	Due	07/10/2026 07:00
	CO3117 - Machine Learning	Format	Formative take-home

Purpose. Build durable foundations for the longitudinal assignment: problem formulation, evaluation discipline, leakage awareness, and model-behaviour reasoning.

- Complete the work by hand where calculation or derivation is requested. Show reasoning; unsupported final numbers receive limited credit.
- Use lecture notes and course resources. Do not use AI, online solution lookup, or classmates' answers for your first attempt. Later supportive AI use must follow the course AI_USE.md protocol.
- These activities are designed for concept transfer and assignment support. They intentionally do not reproduce the wording, ordering, numerical data, or question structure of the midterm.
- No official worked solutions will be released. Students are encouraged to self-check their handwriting with course materials, discussion, and AI assistants after completion.
- Submit one PDF in your course repository and preserve your first attempt. Corrections may be added later without replacing the original submission.

Activity 1 - Frame a learning problem (2.0 pts)

- A postal service wants to route envelopes automatically. Write a compact ML problem specification containing: input, target, decision made from the prediction, one primary metric, and one realistic source of label noise.
- Give one reason why the same raw data might support either a classification task or a regression task depending on the decision objective.

Activity 2 - Evaluation under limited data (2.0 pts)

- A team has 120 labelled examples. Compare these two plans: (i) one 80/20 split; (ii) 5-fold cross-validation on the training portion plus a final untouched test set. Explain what each estimate tells you and what it does not tell you.
- Explain why using the test set repeatedly while choosing hyperparameters makes the final test result optimistic.

Activity 3 - Metrics and decision context (2.0 pts)

- A binary classifier is evaluated on 250 cases: TP=36, FN=14, FP=30, TN=170. Compute accuracy, precision, recall, and F1 for the positive class. Show working.
- The positive class represents a rare but costly event. Give one reason accuracy can be misleading and identify which error type deserves special attention in this context.

Activity 4 - Diagnose model behaviour (2.0 pts)

- Three models have (training error, validation error): A=(24%,26%), B=(3%,21%), C=(9%,11%). Diagnose the likely behaviour of each model and give one concrete intervention for the problematic cases.
- Explain, in words, how model complexity typically changes bias and variance. No formula is required.

Activity 5 - Assignment bridge: protocol card (2.0 pts)

- For the dataset/use case you intend to use in the longitudinal assignment, write a one-page protocol card: frozen target, split policy, primary metric, one leakage risk, one fairness/comparability rule across models, and one limitation you cannot remove by modelling alone.
- This activity is formative: it may be adapted into your assignment repository, but do not copy a generic answer. It must be specific to your own dataset/use case.

Submission checklist

- ☐ Name and student ID on first page
- ☐ All calculations show intermediate steps
- ☐ Assignment-bridge activity uses your own dataset/use case
- ☐ First attempt preserved
- ☐ Any later AI/reference use disclosed according to course policy